STAT/MA 41600
In-Class Problem Set #29: October 26, 2026
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Problem Set 29 Answers

1a. We compute $E(X^2) = \int_0^8 \int_0^{5-5x/8} (x^2)(1/20)\,dy\,dx = 32/3$, or alternatively, $E(X^2) = \int_0^5 \int_0^{8-8y/5} (x^2)(1/20)\,dx\,dy = 32/3$, or alternatively, $E(X^2) = \int_0^8 (x^2)(1/32)(8-x)\,dx = 32/3$. It follows that $\mathrm{Var}(X) = E(X^2) - (E(X))^2 = 32/3 - (8/3)^2 = 32/9$.

1b. We compute $E(Y^2) = \int_0^8 \int_0^{5-5x/8} (y^2)(1/20)\,dy\,dx = 25/6$, or alternatively, $E(Y^2) = \int_0^5 \int_0^{8-8y/5} (y^2)(1/20)\,dx\,dy = 25/6$, or alternatively, $E(Y^2) = \int_0^5 (y^2)(2/25)(5-y)\,dy = 25/6$. It follows that $\mathrm{Var}(Y) = E(Y^2) - (E(Y))^2 = 25/6 - (5/3)^2 = 25/18$.

2a. We compute $E(X^2) = \int_0^\infty \int_0^\infty (x^2)(14e^{-2x-7y})\,dy\,dx = 1/2$, or alternatively, $E(X^2) = \int_0^\infty (x^2)(2e^{-2x})\,dx = 1/2$. It follows that $\mathrm{Var}(X) = E(X^2) - (E(X))^2 = 1/2 - (1/2)^2 = 1/4$.

2b. We compute $E(Y^2) = \int_0^\infty \int_0^\infty (y^2)(14e^{-2x-7y})\,dy\,dx = 2/49$, or alternatively, $E(Y^2) = \int_0^\infty (y^2)(7e^{-7y})\,dy = 2/49$. It follows that $\mathrm{Var}(Y) = E(Y^2) - (E(Y))^2 = 2/49 - (1/7)^2 = 1/49$.

3. We compute $E(X^2) = \int_0^5 \int_0^5 (x^2)(4/1875)(25-xy)\,dy\,dx = 125/18$, or alternatively, $E(X^2) = \int_0^5 (x^2)(4/15 - 2x/75)\,dx = 125/18$. It follows that $\mathrm{Var}(X) = E(X^2) - (E(X))^2 = 125/18 - (20/9)^2 = 325/162$.

4a. We compute $E(X^2) = \int_0^5 \int_0^2 (x^2)(1/25)(5-x)(2-y)\,dy\,dx = 25/6$, or alternatively, $E(X^2) = \int_0^5 (x^2)(2/25)(5-x)\,dx = 25/6$. It follows that $\mathrm{Var}(X) = E(X^2) - (E(X))^2 = 25/6 - (5/3)^2 = 25/18$.

4b. We compute $E(Y^2) = \int_0^5 \int_0^2 (y^2)(1/25)(5-x)(2-y)\,dy\,dx = 2/3$, or alternatively, $E(Y^2) = \int_0^2 (y^2)(1/2)(2-y)\,dy = 2/3$. It follows that $\mathrm{Var}(Y) = E(Y^2) - (E(Y))^2 = 2/3 - (2/3)^2 = 2/9$.

5a. We compute $E(X^2) = \int_0^\infty \int_x^\infty (x^2)(63e^{-2x-7y})\,dy\,dx = 2/81$, or alternatively, $E(X^2) = \int_0^\infty (x^2)(9e^{-9x})\,dx = 2/81$. It follows that $\mathrm{Var}(X) = E(X^2) - (E(X))^2 = 2/81 - (1/9)^2 = 1/81$.

5b. We compute $E(Y^2) = \int_0^\infty \int_x^\infty (y^2)(63e^{-2x-7y})\,dy\,dx = 386/3969$, or alternatively, $E(Y^2) = \int_0^\infty (y^2)(63/2)(e^{-7y} - e^{-9y})\,dy = 386/3969$. It follows that $\mathrm{Var}(Y) = E(Y^2) - (E(Y))^2 = 386/3969 - (16/63)^2 = 130/3969$.